classPPGMSB

A LaTeX 2ε class for theses, dissertations and qualification exams at PPGMSB/UNEB

Classe LaTeX 2ε para teses, dissertações e qualificações — PPGMSB/UNEB

Marcos Batista Figueredo

mbfigueredo@uneb.br | ppgmsb.uneb.br

Version 1.0.0 | December 2025

Abstract. The *classPPGMSB* class provides a complete LaTeX 2ε template for master's dissertations, doctoral theses, and qualification exams submitted to the Graduate Program in Biosystems Modeling and Simulation (PPGMSB) at the State University of Bahia (UNEB), Brazil. Built on the *memoir* class, it automates cover pages, title pages, approval sheets, pre-textual environments, a metadata colophon page with QR codes, bilingual support (Portuguese and English), configurable logos, and SDG icons.

Resumo. A classe *classPPGMSB* fornece um template completo em LaTeX 2ε para dissertações de mestrado, teses de doutorado e exames de qualificação do PPGMSB/UNEB. Baseada na classe *memoir*, automatiza capa, contra-capa, folha de aprovação, elementos pré-textuais, página de metadados com QR code, suporte bilíngue, logos configuráveis e ícones ODS.

Contents

1 Introduction / Introdução 3
- 1.1 License / Licença . . . 3
- 1.2 Requirements / Requisitos . . . 3

2 Installation / Instalação 3

3 Preamble Setup / Configuração do Preâmbulo 4
- 3.1 Abbreviations and Symbols / Abreviações e Símbolos . . . 4
  - 3.1.1 Defining Symbols / Definindo Símbolos . . . 4
  - 3.1.2 Defining Abbreviations / Definindo Abreviações e Siglas . . . 5
  - 3.1.3 Using in Text / Usando no Texto . . . 5
  - 3.1.4 Printing the Lists / Imprimindo as Listas . . . 5
- 3.2 Internal Configuration / Configuração Interna . . . 6

4 Class Options / Opções da Classe 6

5 Document Identification / Identificação do Documento 7
- 5.1 Institution / Instituição . . . 7
- 5.2 Institutional Logos / Logos Institucionais . . . 7
- 5.3 Author / Autor(a) . . . 8

5.4 Title and Subtitle / Título e Subtítulo . . . 8
5.5 Keywords / Palavras-Chave . . . 9
5.6 Advisor / Orientador(a) . . . 9
5.7 Co-advisors / Coorientadores (A, B, C) . . . 9
5.8 Examination Board / Banca Examinadora . . . 10
5.9 Defense Information / Dados da Defesa . . . 11
5.10 Academic Alignment / Aderência Acadêmica . . . 12
5.11 SDGs / Objetivos de Desenvolvimento Sustentável (ODS) . . . 12
5.12 Additional Metadata / Metadados Adicionais . . . 12
6 Document Structure / Estrutura do Documento 13
6.1 Cover, Title Page, Approval Sheet . . . 13
6.2 Acknowledgements / Agradecimentos . . . 14
6.3 Dedication / Dedicatória . . . 14
6.4 Abstract / Resumo e Abstract . . . 14
6.5 Block Quotation / Citação Longa . . . 15
6.6 Pre-textual Lists . . . 15
6.7 Metadata Colophon Page / Página de Metadados . . . 15
6.8 Metadata Fields and Automatic Commands / Campos e Comandos Automáticos . . . 16
7 Figures, Tables, and Algorithms 17
8 Mathematics / Matemática 17
9 Page Style and Colors / Estilo e Cores 18
10 Bibliography / Referências 18
11 Validation Messages / Mensagens de Validação 18
12 Known Limitations / Limitações Conhecidas 19
13 Documentation Glossary and Index / Glossário e Índice da Documentação 19
Glossary / Glossário 19
14 Version History / Histórico de Versões 23
Acknowledgements 23

1 Introduction / Introdução

classPPGMSB was developed to enforce the formatting rules of the PPGMSB graduate program and to free students from the burden of typographic configuration. All structural elements, cover, title page, approval sheet, pre-textual environments, and the metadata colophon, are generated automatically from declarative commands in the document preamble.

> **The class file `classPPGMSB.cls` must never be modified.** All formatting decisions are embedded in the class to guarantee institutional conformity. Altering the class may cause compilation errors or produce documents that do not conform to program standards.
>
> **PT:** O arquivo `classPPGMSB.cls` **nunca deve ser modificado**. Qualquer alteração pode causar erros de compilação ou gerar documentos fora do padrão exigido pelo PPGMSB.

1.1 License / Licença

classPPGMSB is distributed under the **LaTeX Project Public License** (LPPL), version 1.3c or later. The current maintainer is Marcos Batista Figueredo. See `https://www.latex-project.org/lppl.txt`.

1.2 Requirements / Requisitos

- **TEX engine:** pdfLaTeX, XeLaTeX, or LuaLaTeX (TeX Live 2020 or later; MiKTeX 21 or later)
- **Base class:** *memoir*
- **Key packages** (loaded automatically): *iftex*, *etoolbox*, *environ*, *xstring*, *xcolor*, *geometry*, *graphicx*, *tikz*, *algorithm2e*, *babel*, *amsmath*, *pgfplots*, *apacite*, *caption*, *qrcode*, *hyperref*, *glossaries-extra*, *indentfirst*, *lastpage*
- **Template configuration:** the class already loads *hyperref* and *glossaries-extra*; use only `\hypersetup` in the template to configure link colours.

2 Installation / Instalação

1. Download the package from CTAN or from `https://ppgmsb.uneb.br`.
2. Extract the ZIP file; keep the original folder structure intact.
3. Open `dissertacao.tex` and start filling in your data.

> **PT:** Extraia o ZIP numa pasta de trabalho e *não renomeie nem mova* os arquivos ou pastas. A estrutura original é necessária para que o template funcione corretamente.

The expected directory structure:

```
minha-dissertacao/
  classPPGMSB.cls          % class file - DO NOT MODIFY
  dissertacao.tex          % main file - edit this
  figuras/                 % logos and images
    logo_uneb.jpg          % download from ppgmsb.uneb.br (300 dpi)
```

```
    ppgmsbLogo.png          % program logo (300 dpi)
    logoso.png              % header icon (300 dpi)
    ods1.png ... ods17.png % SDG icons (download from un.org)
  preTextual/
    simbolos.tex            % symbols list
    abreviacoes.tex         % abbreviations list
  textual/
    introducao.tex
    revisao.tex
  posTextual/
    referencias.bib
    apendice.tex
```

3 Preamble Setup / Configuração do Preâmbulo

The preamble of `dissertacao.tex` contains two elements that must be preserved exactly: the `\input` of the symbol and abbreviation files, and the `\opcoesDocumento` call. The *glossaries-extra* package and the glossary style are set up by the class.

> **Do not add, remove, or reorder packages in the preamble unless you fully understand the consequences.** The class is designed to load all necessary packages internally. Conflicting packages may cause errors that are difficult to diagnose.
>
> **PT:** Não adicione, remova ou reordene pacotes no preâmbulo sem entender as consequências. A classe carrega internamente o que é necessário.

3.1 Abbreviations and Symbols / Abreviações e Símbolos

Abbreviations, acronyms, and mathematical symbols are managed by the *glossaries-extra* package, which automates their first-use expansion and list generation. The class loads the package, defines the dotted list style, and sorts the lists with LaTeX itself (`\makenoidxglossaries`): no external program is needed. Define all entries in the preamble (the `preTextual` files are read there); only entries used in the text with `\gls` appear in the lists.

3.1.1 Defining Symbols / Definindo Símbolos

Symbols are defined in `preTextual/simbolos.tex` using `\newglossaryentry`:

```
\newglossaryentry{pi}{
  name        = {\ensuremath{\pi}},
  symbol      = {\ensuremath{\pi}},
  description = {Ratio of a circle's circumference to its diameter},
  type        = symbols
}

\newglossaryentry{lambda}{
  name        = {\ensuremath{\lambda}},
  symbol      = {\ensuremath{\lambda}},
  description = {Wavelength in wave theory},
  type        = symbols
}
```

Listing 1: Defining symbols in preTextual/simbolos.tex

The `type=symbols` key is mandatory; it routes the entry to the symbols list rather than the default glossary.

3.1.2 Defining Abbreviations / Definindo Abreviações e Siglas

Abbreviations and acronyms are defined using `\newabbreviation` in `preTextual/abreviacoes.tex`:

```
\newabbreviation{ai}{AI}{Inteligencia Artificial}
\newabbreviation{iot}{IoT}{Internet das Coisas}
\newabbreviation{ppgmsb}{PPGMSB}{Programa de Pos-Graduacao em
    Modelagem e Simulacao de Biossistemas}
```

Listing 2: Defining abbreviations in preTextual/abreviacoes.tex

3.1.3 Using in Text / Usando no Texto

Command	Behaviour
`\gls{ai}`	First use: *Inteligência Artificial* (*AI*); subsequent uses: *AI*
`\gls{pi}`	Inserts the symbol π and registers it in the symbols list
`\glspl{ai}`	Plural form
`\Gls{ai}`	First letter capitalised

3.1.4 Printing the Lists / Imprimindo as Listas

Place these commands in the pre-textual section, after the abstracts:

```
% List of Symbols / Lista de Simbolos
\listaDeSimbolos

% List of Abbreviations / Lista de Abreviacoes
\listaDeAbreviacoes
```

Listing 3: Printing lists — place after abstracts

\listaDeSimbolos **\listaDeSimbolos** **\listaDeAbreviacoes**

Print the list of symbols and the list of abbreviations with the dotted style (*symbol* : *description* ... *pages*). Titles follow the document language: *Lista de Símbolos* and *Lista de Abreviações*, or *List of Symbols* and *List of Abbreviations* with `diss=en`, `tese=en`, or `ing`. If no entry of a list is used in the text, the list is not printed (no title and no blank page).

Entries are sorted alphabetically, word by word and ignoring case (`sort=word`). Accents are ignored when sorting, so *Água* comes between *abacate* and *Casa*, and *Índice* between *ilha* and *IoT*; the printed text keeps its accents. Symbols defined as `\ensuremath{\alpha}` are ordered by the command name (α, λ, π, ε); use the `sort` key of `\newglossaryentry` to choose another order.

Compilation: no extra step is needed. The lists are built by LaTeX alone, with pdfLaTeX, XeLaTeX, or LuaLaTeX, locally, on Overleaf, or on a compilation server, without `makeglossaries`, `makeindex`, `xindy`, `bib2gls`, or shell-escape. As with the table of contents, the lists appear from the second LaTeX run on; `latexmk` and Overleaf rerun automatically:

```
latexmk -pdf dissertacao
```

PT: Não há passo extra. As listas de símbolos e abreviações são geradas só pelo LaTeX (pdfLaTeX, XeLaTeX ou LuaLaTeX), também no Overleaf, sem `makeglossaries` e sem shell-escape. Como o sumário, elas aparecem a partir da segunda compilação; o `latexmk` e o Overleaf recompilam sozinhos.

Older documents: documents created from earlier versions of the template still load *glossaries-extra* with `automake`, call `\makeglossaries`, define the `dotglos-extra` style, and use `\printglossary`. They still compile: `\makeglossaries` is ignored, `\printglossary` works like `\printnoidxglossary`, and each of these commands emits a `\ClassWarning`. To update, delete those lines from the preamble and replace the two `\printglossary` calls with `\listaDeSimbolos` and `\listaDeAbreviacoes`.

PT: Documentos antigos que ainda usam `\makeglossaries` e `\printglossary` compilam normalmente, com avisos. Para atualizar, apague essas linhas do preâmbulo e use `\listaDeSimbolos` e `\listaDeAbreviacoes`.

3.2 Internal Configuration / Configuração Interna

\opcoesDocumento **\opcoesDocumento**

This command applies global configuration based on the chosen class option and activates the symbol and abbreviation lists. It must be called *once*, in the preamble, *after* the `\input` of the symbol and abbreviation files and *before* `\begin{document}`. Do not remove, comment, or duplicate it.

PT: O comando `\opcoesDocumento` deve aparecer exatamente uma vez, no preâmbulo, após o `\input` dos arquivos de símbolos e abreviações e antes de `\begin{document}`. **Não remova, comente ou duplique.**

4 Class Options / Opções da Classe

Load the class with exactly one document-type option:

```
\documentclass[⟨option⟩]{classPPGMSB}
```

Option	Language	Type
`diss`	Portuguese	Master's dissertation — dissertação (default)
`tese`	Portuguese	Doctoral thesis — tese de doutorado
`qlf`	Portuguese	Qualification exam — exame de qualificação
`diss=en`	English	Master's dissertation
`tese=en`	English	Doctoral thesis

Exactly **one** option must be given. Specifying two options or no option raises a `\ClassError` and stops compilation.

Equivalence: `diss=en` is equivalent to the legacy syntax `diss,ing`; `tese=en` is equivalent to `tese,ing`. The `ing` option alone (without a document type) is invalid and causes a `\ClassError`. The combination `qlf,ing` is also invalid: qualification exams are supported only in Portuguese.

PT: Exatamente uma opção deve ser fornecida. Mais de uma ou nenhuma gera `\ClassError`. A sintaxe legada `diss,ing` é equivalente a `diss=en`.

5 Document Identification / Identificação do Documento

All identification commands are placed inside `\begin{document}`, before `\makeCapa`. They must not be removed or left without a value — leave the argument *empty* (`{}`) if a field does not apply.

PT: Todos os comandos de identificação ficam dentro de `\begin{document}`, antes de `\makeCapa`. Nunca remova as chaves `{}`, **não** deixe comandos em aberto, **não** altere os nomes dos comandos. Se um campo não se aplica, deixe o argumento *vazio*: `{}`.

5.1 Institution / Instituição

The institutional block is **pre-filled** in the template and should not be changed in normal circumstances. Changes are required only when the institutional leadership changes and the program secretariat issues an official update.

```
\Universidade{Universidade do Estado da Bahia}
\Departamento{Departamento de Ciencias Exatas e da Terra --- Campus II}
\Programa{Programa de Pos-Graduacao em Modelagem e Simulacao de
    Biossistemas}
\UrlUNEB{https://www.uneb.br}
\UrlMSB{https://ppgmsb.uneb.br}
\ReitorNome{Adriana Marmori Lima}
\ProreitorNome{Elizeu Clementino de Souza}
\DiretorNome{Issac Douglas Moreira}
\CoordenadorNome{Marcos Batista Figueredo}
```

Listing 4: Pre-filled institutional data

Commands `\ReitorNome`, `\ProreitorNome`, `\DiretorNome`, and `\CoordenadorNome` are used only in the metadata page.

5.2 Institutional Logos / Logos Institucionais

`\LogoUniversidade``{path}` **`\LogoPrograma`**`{path}` **`\LogoCabecalho`**`{path}`

Paths to the university logo, program logo, and the small header icon.

```
\LogoUniversidade{figuras/logo_uneb.jpg}
\LogoPrograma{figuras/ppgmsbLogo.png}
\LogoCabecalho{figuras/logoso.png}
```

Listing 5: Logo paths — place files in figuras/

Logos are not included in the CTAN package due to trademark restrictions. Download them from `https://ppgmsb.uneb.br` and place them in the `figuras/` directory.

Image quality: Use files at **300 dpi or higher**. Lower-resolution images produce blurry logos in the printed document.

If a logo file is missing, a cross-box placeholder is displayed and compilation continues normally.

PT: Logos não incluídos (marca registrada). Baixe em `https://ppgmsb.uneb.br`. Use imagens a **300 dpi ou mais**. Arquivo ausente: exibe placeholder, compilação continua normalmente.

5.3 Author / Autor(a)

\NomeAutor **\NomeAutor**{*full name*}

Full name without abbreviations, exactly as in official records.

\EmailAutor **\EmailAutor**{*email*}

Active institutional or personal e-mail.

\LinkLattes **\LinkLattes**{*URL*}

URL of the author's CNPq Lattes curriculum.

Lattes is mandatory in Brazil. The CNPq Lattes curriculum is the official academic record system for Brazilian researchers and graduate students. All PPGMSB students must maintain an up-to-date Lattes curriculum. URL pattern: `http://lattes.cnpq.br/` followed by the 16-digit ID.

PT: O Currículo Lattes é **obrigatório** no Brasil — é o identificador oficial da produção acadêmica nacional. Todos os discentes do PPGMSB devem manter o Lattes atualizado.

\LinkOrcid **\LinkOrcid**{*number only*}

Provide only the **16-digit ORCID number**, not the full URL. Format: `0000-0000-0000-0000`. The class formats and hyperlinks the identifier automatically.

```
\LinkOrcid{0000-0002-1825-0097}   % correct: number only
% NOT: https://orcid.org/0000-0002-1825-0097  (do not include URL)
```

\AutorNacionalidade \AutorMunicipio \AutorEstado **\AutorNacionalidade**{*text*} **\AutorMunicipio**{*city*} **\AutorEstado**{*UF*}

\DataNascimento **\DataNascimento**{*DD/MM/YYYY*}

Nationality, birthplace, and date of birth. Used only in the metadata page for administrative cataloguing.

5.4 Title and Subtitle / Título e Subtítulo

\TituloBR **\TituloBR**{*text*} **\TituloEN**{*text*}

Title in Portuguese and English. Maximum **150 characters** each. If exceeded, the class highlights the overflow in yellow with a red warning.

Subtitle policy at PPGMSB / Política de subtítulo no PPGMSB

The subtitle is **not standard** at PPGMSB. A title should be self-sufficient to describe the work. The use of a subtitle must be explicitly discussed with and approved by the examination board.

If the work has no subtitle — which is the default — **leave the commands with empty arguments**; do not remove or comment them:

```
\SubTituloBR{}   \SubTituloEN{}
```

PT: O subtítulo **não é padrão** no PPGMSB. Seu uso deve ser justificado e aprovado em banca. Sem subtítulo, deixe o campo vazio: `\SubTituloBR{}`.

\SubTituloBR \SubTituloEN

\SubTituloBR{*text*} **\SubTituloEN**{*text*}

Optional subtitles. Maximum **80 characters** each. The class automatically suppresses the subtitle line when the argument is empty.

5.5 Keywords / Palavras-Chave

\palavrasChavePT **\palavrasChavePT**{*kw1; kw2; kw3*}

Keywords in Portuguese, separated by semicolons. The class counts terms using the semicolon separator and emits a `\ClassWarning` when the non-empty list has fewer than 3 or more than 7 terms.

\palavrasChaveEN **\palavrasChaveEN**{*kw1; kw2; kw3*}

Keywords in English, same format and same automatic warning rule.

\palavrasChaveBusca **\palavrasChaveBusca**{*extended terms*}

Extended keyword set for indexing and repository search systems (printed on the metadata page only, not appended to the abstracts). Use it for compound terms, common variants, spelling variants, broader subject terms, and institutional tags such as UNEB and PPGMSB.

5.6 Advisor / Orientador(a)

\OrientadorNome \OrientadorSigla

\OrientadorNome{*name*} **\OrientadorSigla**{*acronym*}

\OrientadorFiliacao **\OrientadorFiliacao**{*institution*}

Advisor's full name, institution acronym, and institution name.

\OrientadorLattes \OrientadorOrcid

\OrientadorLattes{*URL*} **\OrientadorOrcid**{*number only*}

Advisor's Lattes URL and ORCID number (format: `0000-0000-0000-0000`).

5.7 Co-advisors / Coorientadores (A, B, C)

Up to three co-advisors (A, B, C) are supported. **If there is no co-advisor, leave all co-advisor commands with empty arguments — do not comment them out or remove them.**

```
% If no co-advisor, leave all fields empty:
\CoorientadorANome{}
\CoorientadorAFiliacao{}
\CoorientadorASigla{}
\CoorientadorAFormacao{}
\CoorientadorAAtuacao{}
\CoorientadorAOrcid{}
\CoorientadorALattes{}
```

Listing 6: Co-advisor block — fill or leave empty

When a co-advisor exists, fill all fields:

```
\CoorientadorANome{Full Name of Co-advisor A}
\CoorientadorAFiliacao{Institution Name}
```

```
\CoorientadorASigla{ACRONYM}
\CoorientadorAFormacao{Academic background (e.g., Computer Science)}
\CoorientadorAAtuacao{Current research area (e.g., Computational Biology)}
\CoorientadorAOrcid{0000-0000-0000-0000}
\CoorientadorALattes{http://lattes.cnpq.br/0000000000000000}
```

Listing 7: Co-advisor A with data

Substitute `A` with `B` or `C` for additional co-advisors.

Formação vs. Atuação:

- **Formação** (*Academic background*): the main field of the doctoral degree — e.g., "Engenharia Elétrica", "Ciência da Computação".
- **Atuação** (*Research area*): the current research specialisation — e.g., "Modelagem Computacional e Biossistemas".

The same distinction applies to examination board members.

5.8 Examination Board / Banca Examinadora

`\numeroMembrosBanca` **`\numeroMembrosBanca`**`{`*n*`}`

Total number of board members **including the advisor**. Accepted values: 3, 4, 5, or 6.

Regra de composição da banca / Board composition rules:

The advisor is automatically included as president of the board. **Do not list the advisor again** in the member blocks (A–F).

For **master's dissertations**, the standard composition is **3 members**:

- President: the advisor (automatic)
- Member A: 1 internal member (type 1)
- Member B: 1 external member (type 2)

The minimum is always **3 members**; the maximum is **6**. Additional members may be included when the interdisciplinary nature of the research or specific program requirements justify it.

Co-advisors are **not** board members — they do not sign the evaluation record and must not appear in this block.

MembroATipo{1} = internal to the program or institution
MembroATipo{2} = external to the program or institution

Fill only the member blocks that correspond to the declared count. For unused member slots (e.g., D, E, F when count is 3), **leave all commands with empty arguments**:

```
\numeroMembrosBanca{3}

% Member A -- internal
\MembroANome{Prof. Dr. Full Name}
\MembroAFiliacao{Universidade do Estado da Bahia}
\MembroASigla{UNEB}
\MembroAFormacao{Biologia Computacional}
\MembroAAtuacao{Modelagem de Sistemas Complexos}
```

```
\MembroAOrcid{0000-0001-0001-0001}
\MembroALattes{http://lattes.cnpq.br/1111111111111111}
\MembroATipo{1}        % 1 = internal / interno

% Member B -- external
\MembroBNome{Profa. Dra. Full Name}
\MembroBFiliacao{Universidade Federal da Bahia}
\MembroBSigla{UFBA}
\MembroBFormacao{Ciencia da Computacao}
\MembroBAutacao{Inteligencia Artificial}
\MembroBOrcid{0000-0002-0002-0002}
\MembroBLattes{http://lattes.cnpq.br/2222222222222222}
\MembroBTipo{2}        % 2 = external / externo

% Unused members (count = 3): leave empty
\MembroCNome{}\MembroCFiliacao{}\MembroCSigla{}
\MembroCFormacao{}\MembroCAtuacao{}\MembroCOrcid{}
\MembroCLattes{}\MembroCTipo{}
\MembroDNome{}\MembroDFiliacao{}\MembroDSigla{}
\MembroDFormacao{}\MembroDAtuacao{}\MembroDOrcid{}
\MembroDLattes{}\MembroDTipo{}
```

Listing 8: Examination board example: 3 members (master's standard)

Count	Advisor	Required members	Notes
3	Automatic	A (type 1) + B (type 2)	Master's standard
4	Automatic	A + B + C	One extra
5	Automatic	A + B + C + D	Two extra
6	Automatic	A + B + C + D + E	Maximum

5.9 Defense Information / Dados da Defesa

\HoraDefesa **\DataDefesa**{*DD/MM/YYYY*} **\HoraDefesa**{*HH:MMh*} **\LocalDefesa**{*city/UF*}

Date, time, and location of the defense. Displayed on the title page and metadata page.

\UrlTrabalho \Msbdoi **\UrlTrabalho**{*URL*} **\Msbdoi**{*DOI*}

> **These two fields are provided by the program secretariat after the defense.** They must be left blank (empty argument) during the writing phase and the qualification exam. Fill them in only for the **final deposit version** of the document, using the identifiers communicated officially by the secretariat.
>
> **PT:** Esses dois campos são fornecidos pela secretaria do programa **após a defesa**. Devem ficar em branco durante a escrita e a qualificação. Preencha **apenas na versão final de depósito**.

```
\DataDefesa{01/12/2025}
\HoraDefesa{14:00h}
\LocalDefesa{Alagoinhas/BA}
%--- Informed by the secretariat AFTER the defense ---
\UrlTrabalho{}    % fill only in the final deposit version
\Msbdoi{}         % fill only in the final deposit version
```

Listing 9: Defense data — URL and DOI are post-defense

5.10 Academic Alignment / Aderência Acadêmica

\AreaDeConcentracaoBR **\AreaDeConcentracaoBR**{*text*} **\AreaDeConcentracaoEN**{*text*}
Area of concentration in Portuguese and English.

\LinhaPesquisaBR **\LinhaPesquisaBR**{*text*} **\LinhaPesquisaEN**{*text*}
Research line in Portuguese and English.

\GrupoDePesquisaBR **\GrupoDePesquisaBR**{*text*} **\GrupoDePesquisaEN**{*text*}
Research group in Portuguese and English. Indicate only the *primary* research group, exactly as registered in the official CNPq research group directory.

5.11 SDGs / Objetivos de Desenvolvimento Sustentável (ODS)

\odsSelecionadas **\odsSelecionadas**{*n*, *n*, *n*, ...}
Comma-separated list of SDG numbers (1–17) aligned with the work.

```
\odsSelecionadas{3,7,13,15}
```

The selection must reflect *genuine* alignment with the research. Discuss with your advisor before finalising the selection.

File naming / Nomes dos arquivos: The class looks for SDG icon files named `ods`⟨*n*⟩`.png` in the `figuras/` folder, where ⟨*n*⟩ is the SDG number. For example, SDG 3 requires `figuras/ods3.png`, SDG 15 requires `figuras/ods15.png`, and so on up to `ods17.png`. If a file is missing, a numbered coloured placeholder is displayed and compilation continues.

SDG images are not included in the CTAN package. They are subject to UN copyright. Download the official SDG icon set from `https://www.un.org/sustainabledevelopment/news/communications-material/` and save the files as `figuras/ods1.png`, `figuras/ods2.png`, ..., `figuras/ods17.png`.

\ODSFiguras **\ODSFiguras**[*height*]
Renders the selected SDG icons as a horizontal strip. Default height: `2.5cm`. Called automatically by `\paginaMetadados`; can also be used explicitly in the text body.

5.12 Additional Metadata / Metadados Adicionais

\Licenca **\Licenca**[*URL*]{*label*} (default: CC BY-NC-SA 4.0)
License name and URL. Change only with explicit justification (e.g., secrecy agreement, industrial partnership, legal restriction).

\Financiamento **\Financiamento**{*text*}
List all funding sources: scholarships, grants, project numbers. If there is no funding, leave empty `{}`.

\Repositorio **\Repositorio**{*URL*}
URL of a public code or data repository (GitHub, Zenodo, etc.), if applicable. Leave empty otherwise.

\Volumes `\Volumes{n}`

Number of volumes (default: 1).

6 Document Structure / Estrutura do Documento

```
\begin{document}
  % 1. Identification commands (see Section 5)
  \NomeAutor{...}  \TituloBR{...}  ...

  % 2. Pre-textual element generation
  \makeCapa
  \makeContraCapa
  \folhaAprovacao

  % 3. Front matter (roman numbering)
  \frontmatter
  \begin{agradecimentos}...\end{agradecimentos}  % optional; not in qlf
  \begin{dedicar}...\end{dedicar}                % optional; not in qlf
  \begin{resumo}...\end{resumo}
  \begin{newAbstract}...\end{newAbstract}
  \listaDeSimbolos
  \listaDeAbreviacoes
  \listaDeIlustracoes
  \listaDeTabelas
  \listaDeAlgoritmos
  \tableofcontents*

  % 4. Main text
  \mainmatter
  \pagestyle{ppgmsbPages}
  \include{textual/introducao}
  \include{textual/revisao}
  ...

  % 5. Post-textual
  \bibliography{posTextual/referencias}
  \paginaMetadados

\end{document}
```

Listing 10: Recommended structure / Estrutura recomendada

6.1 Cover, Title Page, Approval Sheet

\makeCapa **`\makeCapa`**

Generates the cover (capa) with title, author, advisor, and logos.

\makeContraCapa **`\makeContraCapa`**

Generates the title page (contra-capa) with institutional information, nature-of-work text, defense data, and optional DOI.

\folhaAprovacao **\folhaAprovacao**

Generates the approval sheet with signature boxes, sized automatically according to \numeroMembro

6.2 Acknowledgements / Agradecimentos

Acknowledgements (agradecimentos) and dedication (dedicar) are pre-textual elements that belong to the **final defense document** (options diss, tese). **They should not appear in the qualification exam document** (qlf). Comment them out when working on the qualification version.

PT: Agradecimentos e dedicatória são elementos da **defesa final** e **não devem constar na qualificação** (qlf). Comente-os ao trabalhar na versão de qualificação.

agradecimentos environment

Acknowledgements section. Text is automatically truncated at **1000 characters** with a red warning if exceeded. When the student holds a scholarship, the funding agency acknowledgement is **mandatory**. Recommended length: 700–1000 characters (approximately 100–150 words).

```
\begin{agradecimentos}
Agradeço à Universidade do Estado da Bahia e ao PPGMSB pela oportunidade.
Sou grato(a) ao(à) orientador(a) [Nome], pela orientação dedicada.
Agradeço aos professores e colegas de curso pelo convívio e trocas. À minha
família, pelo apoio incondicional. O(a) autor(a) agradece à CAPES pela
concessão da bolsa de estudos.
\end{agradecimentos}
```

Listing 11: Acknowledgements environment

6.3 Dedication / Dedicatória

dedicar environment

Brief dedication page (optional; not in qlf). Right-aligned in the lower third of the page. Maximum **400 characters** (enforced by the class with a warning).

```
\begin{dedicar}
  Dedico este trabalho aos meus filhos e à minha família,
  razão de cada passo desta jornada.
\end{dedicar}
```

Listing 12: Dedication environment

6.4 Abstract / Resumo e Abstract

resumo environment

Abstract in Portuguese. Keywords are appended automatically from \palavrasChavePT.

newAbstract environment

Abstract in English. Keywords from \palavrasChaveEN are appended.

Word limit / Limite de palavras: PPGMSB norms recommend **250 to 300 words** per abstract. Write in a single paragraph, using impersonal language, covering: objectives, methodology, main results, and conclusions. Do not include citations, references, or enumerations.

Unlike the agradecimentos and dedicar environments, the class does *not* truncate the abstract automatically — respecting the word limit is the student's responsibility.

PT: A norma do PPGMSB recomenda **250 a 300 palavras** por resumo/abstract, em parágrafo único, sem subdivisões ou referências. A classe *não* trunca automaticamente — cumprir o limite é responsabilidade do discente.

```
\begin{resumo}
  % 250-300 words, single paragraph, impersonal language.
  % Keywords are added automatically; do not insert them manually.
  Este trabalho propoe...
\end{resumo}

\begin{newAbstract}
  % 250-300 words, single paragraph.
  This work proposes...
\end{newAbstract}
```

Listing 13: Abstract environments

Note for English documents: When using `diss=en` or `tese=en`, the `resumo` environment should be omitted; include only `newAbstract`.

6.5 Block Quotation / Citação Longa

`citacao` environment

Block quotation with 4 cm left indent and single spacing. Required by ABNT for direct quotations of 40 words or more.

6.6 Pre-textual Lists

\listaDeIlustracoes \listaDeTabelas \listaDeAlgoritmos

`\listaDeIlustracoes` `\listaDeTabelas` `\listaDeAlgoritmos`

Generate the lists of figures, tables, and algorithms, respectively. Titles adapt automatically to the document language. Each command also clears to a new page.

6.7 Metadata Colophon Page / Página de Metadados

\paginaMetadados

`\paginaMetadados`

Generates the metadata colophon page at the end of the document.

Why a metadata page? / Por que uma página de metadados?

The *classPPGMSB* metadata colophon is an **innovation in Brazilian academic publishing**. While traditional theses and dissertations end after the references or appendices, the PPGMSB requires a final structured page that consolidates all identifying information about the work and its production context.

This page serves multiple purposes:

- **Discoverability:** The QR code links directly to the deposited work in institutional repositories, enabling immediate access from a printed copy.
- **Traceability:** Authorship, advisory, examination board, institutional affiliation, and academic identifiers (Lattes, ORCID, DOI) are consolidated in one verifiable location.

- **Statistical transparency:** Automatically computed counters (pages, figures, tables, equations, references, algorithms) provide a concise profile of the work.
- **SDG alignment:** The selected SDG icons make the sustainability commitments of the work visible and auditable.
- **Institutional compliance:** The license, funding, and repository fields satisfy requirements of funding agencies, publishers, and open-science mandates.

PT: A página de metadados é uma **inovação do PPGMSB** para dissertações no Brasil. Ela consolida autoria, orientação, banca, identificadores acadêmicos, estatísticas do documento, ícones ODS, QR code e licença — informações essenciais para a rastreabilidade e indexação do trabalho em repositórios e bases de dados acadêmicas. Ela complementa e enriquece a contra-capa e a folha de aprovação com dados verificáveis e permanentes.

6.8 Metadata Fields and Automatic Commands / Campos e Comandos Automáticos

metadata page generation is based on the identification commands filled in before `\makeCapa`. The command `\paginaMetadados` reads these stored values and prints them in a final colophon page.

Field group	Commands used by `\paginaMetadados`
Institution	`\Universidade`, `\Departamento`, `\Programa`, `\UrlUNEB`, `\UrlMSB`, `\ReitorNome`, `\ProreitorNome`, `\DiretorNome`, `\CoordenadorNome`
Author	`\NomeAutor`, `\EmailAutor`, `\AutorNacionalidade`, `\AutorMunicipio`, `\AutorEstado`, `\DataNascimento`, `\LinkOrcid`, `\LinkLattes`
Academic alignment	`\AreaDeConcentracaoBR`, `\AreaDeConcentracaoEN`, `\LinhaPesquisaBR`, `\LinhaPesquisaEN`, `\GrupoDePesquisaBR`, `\GrupoDePesquisaEN`
Defense and identifiers	`\DataDefesa`, `\HoraDefesa`, `\LocalDefesa`, `\UrlTrabalho`, `\Msbdoi`
Keywords and discovery	`\palavrasChavePT`, `\palavrasChaveEN`, `\palavrasChaveBusca`, `\odsSelecionadas`
Open science	`\Licenca`, `\Financiamento`, `\Repositorio`, `\Volumes`

The following commands are mostly *read-only helpers*: the class uses them internally, and users normally do not need to call them in the body of the dissertation. They are documented here because they appear in the metadata page implementation and may be useful for diagnostics.

`\QRTrabalho[height]` Generates a QR code from `\UrlTrabalho`. If the URL is empty, it prints a short "URL not defined" message instead of failing.

`\QRDOI[height]` Generates a QR code from `\Msbdoi`, using the `https://doi.org/` resolver. It is available even when the standard metadata page only prints the work URL QR code.

`\ContarFiguras`, `\ContarTabelas`, `\ContarEquacoes`, `\ContarAlgoritmos`, `\ContarReferencias` Print automatic counter values collected during compilation. These values are shown on `\paginaMetadados`. They may need the final LaTeX pass to stabilize.

\motorTeX Prints the detected TeX engine: pdfLaTeX, XeLaTeX, or LuaLaTeX.

\versaoClasse Prints the class version string, currently `1.0.0`.

\dataCompilacao and \horaFormatada Print the compilation date and time. These are provenance fields for the metadata page; they change whenever the PDF is rebuilt.

> **PT:** Esses comandos não são campos para o estudante preencher. Eles consultam dados já informados ou calculados automaticamente pela classe. Em uso normal, basta preencher os comandos declarativos do início do documento e chamar `\paginaMetadados` no final.

7 Figures, Tables, and Algorithms

\fonte **\fonte**{*text*}
Source caption below a figure or table. If empty, prints *Próprio autor* (Author's own).

\nota **\nota**{*text*}
Note below a table (placed after `\fonte`).

```
\begin{figure}[h]
  \centering
  \caption{Title of the figure}
  \includegraphics[width=0.8\textwidth]{figuras/diagram}
  \fonte{\citep{Smith2020}}   % or \fonte{} for own authorship
\end{figure}
```

Listing 14: Figure with source

```
\begin{table}[h]
  \caption{Results}
  \begin{tabular}{lr} ... \end{tabular}
  \fonte{Adapted from \citep{Jones2021}}
  \nota{Values represent mean +/- standard deviation. n=30.}
\end{table}
```

Listing 15: Table with source and note

8 Mathematics / Matemática

The class loads *amsmath*, *amssymb*, and *amsfonts* and sets `\everymath{\displaystyle}` so all inline mathematics uses display style by default.

The following Portuguese-notation operators are predefined:

Command	Output	Command	Output
`\sen`	sen	`\senh`	senh
`\tg`	tg	`\tgh`	tgh
`\cossec`	cossec	`\cotg`	cotg
`\cossech`	cossech	`\cotgh`	cotgh
`\arcsen`	arcsen		

\chaves **\chaves**{*arg*}

Typesets ⟨*arg*⟩ in automatic-size curly braces.

9 Page Style and Colors / Estilo e Cores

Activate the main page style after `\mainmatter`:

```
\mainmatter
\pagestyle{ppgmsbPages}   % header: logo + chapter name; footer: page number
```

The three predefined PPGMSB colours:

Colour name	Hex	Preview	Typical use
`ppgmsbC1`	`#00303A`	dark teal	Rules, boxes, chapter headings
`ppgmsbC2`	`#266054`	green	Section headings, links
`ppgmsbC3`	`#19A340`	bright green	Accents, SDG placeholders

10 Bibliography / Referências

The class uses *apacite* with *natbib* compatibility for APA-style references. Bibliographies are processed with BibTeX. The package includes `brazilian.apc` so *apacite* can load Brazilian Portuguese bibliography terms without an external language file.

`\cite{key}` Direct citation: Author (Year)

`\citep{key}` Indirect citation: (Author, Year)

Place entries in `posTextual/referencias.bib` and call:

```
\bibliography{posTextual/referencias}
```

A complete local compilation sequence for a document with references, glossaries, and cross-references is:

```
xelatex dissertacao
bibtex dissertacao
xelatex dissertacao
xelatex dissertacao
```

or simply `latexmk -pdfxe dissertacao` (`-pdf` for pdfLaTeX, `-pdflua` for LuaLaTeX). The symbol and abbreviation lists need no extra program.

11 Validation Messages / Mensagens de Validação

Title overflow Text exceeding 150 chars (title) or 80 chars (subtitle) is highlighted in yellow with a red counter.

Missing type option No class option → `\ClassError` before compilation.

Multiple type options More than one of `diss`/`tese`/`qlf` → `\ClassError`.

English qualification `qlf,ing` or any English qualification combination → `\ClassError`.

Invalid board count `\numeroMembrosBanca{n}` with $n \notin \{3, 4, 5, 6\}$ → `\PackageError`.

Agradecimentos overflow Text > 1000 chars → truncated with red warning.

Dedicar overflow Text > 400 chars → truncated with red warning.

Keyword count Non-empty `\palavrasChavePT` or `\palavrasChaveEN` with fewer than 3 or more than 7 semicolon-separated terms → `\ClassWarning`.

12 Known Limitations / Limitações Conhecidas

- The class defines setter commands for a sixth explicit examiner group (`\MembroFNome`, `\MembroFFiliacao`, etc.), but the approval sheet currently supports `\numeroMembrosBanca{6}` as advisor plus members A–E. Treat the MembroF commands as reserved/internal until the approval-sheet layout is expanded.

13 Documentation Glossary and Index / Glossário e Índice da Documentação

This documentation includes a glossary and an index, following the common style of reference manuals distributed through CTAN. The command index is generated automatically from documented command signatures introduced with `\cmd`. It also includes a separate *template commands from dissertacao.tex* group with every control sequence found in the distributed template, including standard LaTeX commands and commands that occur in commented examples.

To rebuild these auxiliary lists locally, compile the documentation with:

```
xelatex classPPGMSB-doc
makeindex classPPGMSB-doc
xelatex classPPGMSB-doc
xelatex classPPGMSB-doc
```

The glossary is sorted by LaTeX itself; only the index uses `makeindex`.

Glossary / Glossário

BibTeX Bibliography processor used with *apacite* to generate the reference list.

counter LaTeX register used by the class to count figures, tables, equations, algorithms, references, and pages.

CTAN Comprehensive TeX Archive Network, the main public archive for TeX and LaTeX packages.

glossary Alphabetical or classified list of terms, abbreviations, or symbols generated from marked entries.

index Back-of-document list of commands and topics with page references.

metadata page Final structured page generated by `\paginaMetadados` with identifiers, document statistics, QR codes, licensing, funding, repository, and SDG information.

QR code Two-dimensional code generated from the document URL or DOI to provide quick access from printed copies.

SDG/ODS Sustainable Development Goals / Objetivos de Desenvolvimento Sustentavel represented by numbered icons from 1 to 17.

TeX engine Program used to compile the document, such as pdfLaTeX, XeLaTeX, or LuaLaTeX.

Command and Topic Index / Indice Remissivo

commands

\AreaDeConcentracaoBR, 12
\AreaDeConcentracaoEN, 12
\AutorEstado, 8
\AutorMunicipio, 8
\AutorNacionalidade, 8
\DataDefesa, 11
\DataNascimento, 8
\EmailAutor, 8
\Financiamento, 12
\GrupoDePesquisaBR, 12
\GrupoDePesquisaEN, 12
\HoraDefesa, 11
\Licenca, 12
\LinhaPesquisaBR, 12
\LinhaPesquisaEN, 12
\LinkLattes, 8
\LinkOrcid, 8
\LocalDefesa, 11
\LogoCabecalho, 7
\LogoPrograma, 7
\LogoUniversidade, 7
\Msbdoi, 11
\NomeAutor, 8
\ODSFiguras, 12
\OrientadorFiliacao, 9
\OrientadorLattes, 9
\OrientadorNome, 9
\OrientadorOrcid, 9
\OrientadorSigla, 9
\Repositorio, 12
\SubTituloBR, 9
\SubTituloEN, 9
\TituloBR, 8
\TituloEN, 8
\UrlTrabalho, 11
\Volumes, 13
\chaves, 18
\folhaAprovacao, 13
\fonte, 17
\listaDeAbreviacoes, 5
\listaDeAlgoritmos, 15
\listaDeIlustracoes, 15
\listaDeSimbolos, 5
\listaDeTabelas, 15
\makeCapa, 13
\makeContraCapa, 13
\nota, 17
\numeroMembrosBanca, 10
\odsSelecionadas, 12
\opcoesDocumento, 6
\paginaMetadados, 15
\palavrasChaveBusca, 9
\palavrasChaveEN, 9
\palavrasChavePT, 9

template commands from dissertacao.tex

\AreaDeConcentracaoBR, 19
\AreaDeConcentracaoEN, 19
\AutorEstado, 19
\AutorMunicipio, 19
\AutorNacionalidade, 19
\CoordenadorNome, 19
\CoorientadorAAtuacao, 19
\CoorientadorAFiliacao, 19
\CoorientadorAFormacao, 19
\CoorientadorALattes, 19
\CoorientadorANome, 19
\CoorientadorAOrcid, 19
\CoorientadorASigla, 19
\CoorientadorBAtuacao, 19
\CoorientadorBFiliacao, 19
\CoorientadorBFormacao, 19
\CoorientadorBLattes, 19
\CoorientadorBNome, 19
\CoorientadorBOrcid, 19
\CoorientadorBSigla, 19
\CoorientadorCAtuacao, 19
\CoorientadorCFiliacao, 19
\CoorientadorCFormacao, 19
\CoorientadorCLattes, 19
\CoorientadorCNome, 19
\CoorientadorCOrcid, 19
\CoorientadorCSigla, 19
\DataDefesa, 19
\DataNascimento, 19

\Departamento, 19
\DiretorNome, 19
\EmailAutor, 19
\Financiamento, 19
\GrupoDePesquisaBR, 19
\GrupoDePesquisaEN, 19
\HoraDefesa, 19
\Licenca, 19
\LinhaPesquisaBR, 19
\LinhaPesquisaEN, 19
\LinkLattes, 19
\LinkOrcid, 19
\LocalDefesa, 19
\LogoCabecalho, 19
\LogoPrograma, 19
\LogoUniversidade, 19
\MembroAAtuacao, 19
\MembroAFiliacao, 19
\MembroAFormacao, 19
\MembroALattes, 19
\MembroANome, 19
\MembroAOrcid, 19
\MembroASigla, 19
\MembroATipo, 19
\MembroBAtuacao, 19
\MembroBFiliacao, 19
\MembroBFormacao, 19
\MembroBLattes, 19
\MembroBNome, 19
\MembroBOrcid, 19
\MembroBSigla, 19
\MembroBTipo, 19
\MembroCAtuacao, 19
\MembroCFiliacao, 19
\MembroCFormacao, 19
\MembroCLattes, 19
\MembroCNome, 19
\MembroCOrcid, 19
\MembroCSigla, 19
\MembroCTipo, 19
\MembroDAtuacao, 19
\MembroDFiliacao, 19
\MembroDFormacao, 19
\MembroDLattes, 19
\MembroDNome, 19
\MembroDOrcid, 19
\MembroDSigla, 19
\MembroDTipo, 19
\MembroEAtuacao, 19
\MembroEFiliacao, 19
\MembroEFormacao, 19
\MembroELattes, 19
\MembroENome, 19
\MembroEOrcid, 19
\MembroESigla, 19
\MembroETipo, 19
\Msbdoi, 19
\NomeAutor, 19
\OrientadorFiliacao, 19
\OrientadorLattes, 19
\OrientadorNome, 19
\OrientadorOrcid, 19
\OrientadorSigla, 19
\Programa, 19
\ProreitorNome, 19
\ReitorNome, 19
\Repositorio, 19
\SubTituloBR, 19
\SubTituloEN, 19
\TituloBR, 19
\TituloEN, 19
\Universidade, 19
\UrlMSB, 19
\UrlTrabalho, 19
\UrlUNEB, 19
\Volumes, 19
\addcontentsline, 19
\appendixname, 19
\appendixpagename, 19
\beforechapskip, 19
\begin, 19
\bibliography, 19
\caption, 19
\centering, 19
\cite, 19
\citep, 19
\clearpage, 19
\documentclass, 19
\end, 19
\folhaAprovacao, 19
\fonte, 19
\frontmatter, 19

\gls, 19
\hypersetup, 19
\include, 19
\includegraphics, 19
\input, 19
\listaDeAbreviacoes, 19
\listaDeAlgoritmos, 19
\listaDeIlustracoes, 19
\listaDeSimbolos, 19
\listaDeTabelas, 19
\mainmatter, 19
\makeCapa, 19
\makeContraCapa, 19
\makeglossaries, 19
\numeroMembrosBanca, 19
\odsSelecionadas, 19
\opcoesDocumento, 19
\pagestyle, 19
\paginaMetadados, 19
\palavrasChaveBusca, 19
\palavrasChaveEN, 19
\palavrasChavePT, 19
\pi, 19
\pretextual, 19
\printglossary, 19
\renewcommand, 19
\restoreapp, 19
\setcounter, 19
\setlength, 19
\tableofcontents, 19
\textbf, 19
\textwidth, 19
\usepackage, 19

14 Version History / Histórico de Versões

1.0.0 (2025/12/01) Initial public release on CTAN. Configurable logo paths with placeholder fallback; SDG icon support (`ods⟨n⟩.png`); metadata colophon page with QR code, counters, and SDG strip; character-limit validation for titles, agradecimentos, and dedicatória; bilingual document support (`diss=en`, `tese=en`).

Acknowledgements

The author thanks the students and faculty of PPGMSB/UNEB for feedback that shaped this class, the working group on academic norms for the institutional guidelines documented here, and the TEX community for the outstanding packages on which *classPPGMSB* depends.

O autor agradece aos estudantes e docentes do PPGMSB/UNEB pelas sugestões que moldaram esta classe; ao Grupo de Trabalho de Normas pelas diretrizes aqui documentadas; e à comunidade TEX pelos excelentes pacotes nos quais a *classPPGMSB* se apoia.